Overview of Adversarial examples

Subject: Adversarial examples

1.

argmax k = 1 , . . . , K f k (x ^) ≠ y , | | x ^ − x | | p ≤ ■ and x ^ ∈ [0 , 1] d {\displaystyle {\text{argmax}}_{k=1,...,K}f_{k}({\hat {x}})\neq y,||{\hat {x}}-x||_{p}\leq \epsilon {\text{ and }}{\hat {x}}\in [0,1]^{d}}

2.

Challenges in applying machine learning to security Researchers have observed that the constraints under which machine-learning techniques function in the security domain are different from those of common benchmark domains. Security data may change over time, include mislabeled samples, or reflect adversarial behavior, which complicates evaluation and reproducibility.

3.

Concept drift Because malware creators continuously adapt their techniques, the statistical properties of malicious samples also change. This form of concept drift has been widely documented and may reduce model performance unless systems are updated regularly or incorporate mechanisms for incremental learning.

4.

Data collection issues Security datasets vary across formats, including binaries, network traces, and log files. Studies have reported that the process of converting these sources into features can introduce bias or inconsistencies. In addition, time-based leakage can occur when related malware samples are not properly separated across training and testing splits, which may lead to overly optimistic results.

5.

Feature robustness Researchers differentiate between features that can be easily manipulated and those that are more resistant to modification. For example, simple static attributes, such as header fields, may be altered by attackers, while structural features, such as control-flow graphs, are generally more stable but computationally expensive to extract.

6.

Fast gradient sign method One of the first proposed attacks for generating adversarial examples was proposed by Google researchers Ian J. Goodfellow, Jonathon Shlens, and Christian Szegedy. The attack was called fast gradient sign method (FGSM), and it consists of adding a linear amount of in-perceivable noise to the image and causing a model to incorrectly classify it. This noise is calculated by multiplying the sign of the gradient with respect to the image we want to perturb by a small constant epsilon. As epsilon increases, the model is more likely to be fooled, but the perturbations become easier to identify as well. Shown below is the equation to generate an adversarial example where x {\textstyle x} is the original image, ■ {\textstyle \epsilon } is a very small number, Δ x {\textstyle \Delta _{x}} is the gradient function, J {\textstyle J} is the loss function, θ {\textstyle \theta } is the model weights, and y {\textstyle y} is the true label.

7.

Adversarial attacks and training in linear models There is a growing literature about adversarial attacks in linear models. Indeed, since the seminal work from Goodfellow at al. studying these models in linear models has been an important tool to understand how adversarial attacks affect machine learning models. The analysis of these models is simplified because the computation of adversarial attacks can be simplified in linear regression and classification problems. Moreover, adversarial training is convex in this case. Linear models allow for analytical analysis while still reproducing phenomena observed in state-of-the-art models. One prime example of that is how this model can be used to explain the trade-off between robustness and accuracy. Diverse work indeed provides analysis of adversarial attacks in linear models, including asymptotic analysis for classification and for linear regression. And, finite-sample analysis based on Rademacher complexity. A result from studying adversarial attacks in linear models is that it closely relates to regularization. Under certain conditions, it has been shown that

8.

min (| | δ | | p) subject to C (x + δ) = t , x + δ ∈ [0 , 1] n {\displaystyle \min(||\delta ||_{p}){\text{ subject to }}C(x+\delta)=t,x+\delta \in [0,1]^{n}}

9.

Adversarial examples An adversarial example refers to specially crafted input that is designed to look "normal" to humans but causes misclassification to a machine learning model. Often, a form of specially designed "noise" is used to elicit the misclassifications. Below are some current techniques for generating adversarial examples in the literature (by no means an exhaustive list).

10.

min (| | δ | | p) subject to f (x + δ) ≤ 0 , x + δ ∈ [0 , 1] n {\displaystyle \min(||\delta ||_{p}){\text{ subject to }}f(x+\delta)\leq 0,x+\delta \in [0,1]^{n}}

11.

min (| | δ | | p + c ⋅ f (x + δ)) , x + δ ∈ [0 , 1] n {\displaystyle \min(||\delta ||_{p}+c\cdot f(x+\delta)),x+\delta \in [0,1]^{n}}

12.

System design and learning Feature engineering and training can causes issues. Data snooping is a common pitfall where a model is trained using information that would not be available in a real-world scenario. Spurious correlations result when a model learns to associate artifacts with a label, rather than the underlying security-relevant pattern. For example, a malware classifier might learn to identify a specific compiler artifact instead of malicious behavior itself. Biased parameter selection is a form of data snooping where model hyperparameters are tuned using the test set.

13.

Model extraction Model extraction involves an adversary probing a black box machine learning system in order to extract the data it was trained on. This can cause issues when either the training data or the model itself is sensitive and confidential. For example, model extraction could be used to extract a proprietary stock trading model which the adversary could then use for their own financial benefit. In the extreme case, model extraction can lead to model stealing, which corresponds to extracting a sufficient amount of data from the model to enable the complete reconstruction of the model. On the other hand, membership inference is a targeted model extraction attack, which infers the owner of a data point, often by leveraging the overfitting resulting from poor machine learning practices. Concerningly, this is sometimes achievable even without knowledge or access to a target model's parameters, raising security concerns for models trained on sensitive data, including but not limited to medical records and/or personally identifiable information. With the emergence of transfer learning and public accessibility of many state of the art machine learning models, tech companies are increasingly drawn to create models based on public ones, giving attackers freely accessible information to the structure and type of model being used.